

Resultant force = ma

$$105.8 = (120/9.81) a \Rightarrow a = 8.65 \text{ ms}^{-2}$$

- 11 A** Gravitational potential increases towards infinity. -20 MJ kg^{-1} is at a higher point than -30 MJ kg^{-1} . So mass is moving away from Earth.

$$\text{change in GPE} = \text{mass} \times \text{change in potential} = 1500(-20+30) \times 10^6 = 15 \text{ GJ}$$

- 12 C** From graph, time to melt completely = $11 \times 60 \text{ s}$